

Random Generation of 2-Connected Graphs

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Abstract

Enumerating combinatorial objects is an important research topic in combinatorics. Algorithms for random generating combinatorial objects are called sampling. Sampling has many applications in computer science. We describe a simple algorithm to generate 2-connected graphs based on Markov Chain Monte Carlo method. This algorithm generates graphs at approximately uniform distribution, and we prove it is rapidly mixing.

1 Introduction

Combinatorial objects can sometimes be investigated through random generation in order to discover various properties, estimate statistics or test conjectures. Also objects might be generated randomly to test the behavior or performance of algorithms.

Definiton 1 *An algorithm A is a **random generation algorithm** for a sample space S , if outputs of A is one element of S . Each element's output probability is a distribution π over S . If the distribution π is uniform distribution, then the algorithm is called **uniform random generation algorithm**.*

When considering random generation of graphs, there are many ways to define the sample space. The 2-connected graph is a simple and common graph class which we will define later.

Problem definition: *Given a positive integer n . Design a uniform random generation algorithm for 2-connected simple graph with vertex size n . All isomorphic graphs are considered different.*

1.1 Combinatorial Generation

Enumerative combinatorics is a subfield of combinatorics. The basic problem of enumerative combinatorics is that of counting the number of elements of a finite set[12]. Combinatorial generation is a topic of combinatorial algorithms. Besides counting the number of combinatorial objects, combinatorial generation needs an algorithmic approach to list all objects. The four basic problems is how to list, rank, unrank, random select of objects[1]. Random selection is often called sampling.

Traditionally speaking, if we can enumerative all objects, we also can random sample the objects. Because we can pick a random number between 1 and the numbers of objects, then use unrank algorithm to find the sample object. But this method has some drawbacks. Since the number of objects may be very huge. It may be time consuming in order to unrank the random number. Hence scientist must develop efficient algorithm to sample objects uniformly at random.

1.2 Markov Chain Monte Carlo Method

However other graph classes are difficult to sample, one may use the Markov Chain Monte Carlo method (MCMC). If we want to sample a set of objects, we first devise a Markov chain which state space is these objects. If this Markov chain's stationary distribution is uniform distribution. We can simulate the random walk on it. After adequate many steps, the current state has approximately uniform distribution on state space. Finally we report the object which is represented by current state. The MCMC method can approximate random sampling[10].

Although the Markov chain with uniform stationary distribution can be constructed easily. Theoretically speaking, it may need infinity transitions to achieve stationary distribution. It is

definitely impractical for real world application. Hence the mathematicians define the mixing-time to be the time that Markov chain approximates the stationary distribution. If the mixing-time meets some criterions, we can call it rapidly mixing which is practical for using.

The MCMC method for random generation is firstly published in [3] and the rapidly mixing property is defined in [11]. Over the last 15 years, many mathematical techniques to bind the convergence rate of Markov chain are founded. The paper[9] written by Randall is a good survey and tutorial. As long as we have an approximate sampling algorithm, we also can design an approximate counting algorithm. Exact combinatorial counting is known to be extremely difficult.

1.3 Previous Work

Algorithms for sampling elementary combinatorial objects like permutations and combinations have been designed. But some combinatorial structures are too complicated to sample, some graph classes are kinds of these structures. Random selection algorithm for labeled planar graph and regular graph can be found in these papers[6].

Several graph classes have random generation algorithm based on MCMC method, including bipartite graphs[4], directed acyclic connected graph[7], planar graph[6]. But most of their mixing-times are not proved, it is the hardest part for using MCMC method.

1.4 Contribution

We design a uniform random generation algorithm for 2-connected graph based on MCMC method. After analyzing the mixing-time carefully with some basic mathematical tools, we prove it is rapidly mixing.

1.5 Organization

Section 2 contains some preliminaries, including a formal definition of problem and basic mathematical background for Markov Chain Monte Carlo method. In section 3 we describe the algorithm for generate 2-connected graph and prove it can generate uniform at random. In section 4 we analyze the mixing-time and prove this algorithm is almost rapidly mixing. Section 5 is the conclusion and future work.

2 Preliminaries

Before we exhibit the algorithm, some mathematical backgrounds are needed. In this section, we give a brief introduction of Markov Chain Monte Carlo method. We will also define the mixing-time and list two main techniques for analyzing mixing-time.

2.1 Graph Theory

The knowledge of graph theory can be found in the classic graph theory textbook, we adopt the definition of the textbook[13] in this paper.

2.2 Random Graph

A random graph is a graph that is generated by some random process. The following definition and theorem comes from [?]. Let $G(n, p)$ be the probability space with all n -vertices graphs with edge are chosen independently with probability p . $G(n, 0.5)$ is exactly all n -vertices graphs with the same probability, denoted by $G_{0.5}$.

2.3 Markov Chain

Markov chain is a basic model of stochastic process. The definition below comes from the book[8].

Definiton 2 A *stochastic process* $X = \{X(t) : t \in T\}$ is a collection of random variables over state space S . The index t often represents time, and in that case the process X models the value of a random variable X that changes over time.

We call $X(t)$ the state of the process at time t , X_t for abbreviation. If for all t , X_t assumes values from a countably infinite set, then we say that X is a **discrete space** process. If X_t assumes values from a finite set then the process is **finite**. If T is a countably infinite set we say that X is a **discrete time** process.

Definiton 3 A *discrete time stochastic process* $X_0, X_1, X_2, \dots$ is a **Markov chain** if

$$\begin{aligned} \Pr(X_t = a_t | X_{t-1} = a_{t-1}, X_{t-2} = a_{t-2}, \dots, X_0 = a_0) \\ = \Pr(X_t = a_t | X_{t-1} = a_{t-1}) \end{aligned}$$

Intuitively, the state X_t depends on the previous state X_{t-1} but is independent of the particular history of how the process arrived at state X_{t-1} . This is called **memoryless property** or **Markov property**. We use the notation $P_{i,j}^t$ to represent $\Pr(X_t = j | X_0 = i)$.

Definiton 4 A Markov chain is **ergodic** if it is irreducible and aperiodic. **Irreducible** means that $\forall i, j \in S$, there is a t such that $P_{i,j}^t > 0$. **Aperiodic** means that $\forall i, j \in S$, $\gcd\{t : P_{i,j}^t > 0\} = 1$.

Lemma 1 Any finite, ergodic Markov chain has a unique **stationary distribution** π , such that $\forall i, j \in S \lim_{t \rightarrow \infty} P_{i,j}^t = \pi(j)$.

If a Markov chain reaches a stationary distribution then it maintains the distribution for all future time. Hence the stationary distribution represents a steady state in the chain's behavior. And we know any finite, irreducible and aperiodic Markov chain has a unique stationary distribution.

2.4 Markov Chain Monte Carlo Method

Markov Chain Monte Carlo (MCMC) method is a very general sampling approach to sample from a desired probability distribution. The main idea is to define an irreducible and aperiodic Markov chain whose set of states is the sample space. And the stationary distribution is the desired probability distribution.

If we want to generate a sample, we can simulate the Markov chain, let $X_0, X_1, \dots, X_n$ be a run of the chain. Since the chain converges to stationary distribution from any start state. After a sufficiently larger number of steps T , the distribution X_n is similar to the stationary distribution. So X_T can be used as a sample.

The first step to use MCMC method is to design an irreducible and aperiodic Markov chain. Let the set of states that can transit from x be $N(x)$. $N(x)$ should be a small set such that we can get $N(x)$ by simple computation. We only need to prove the chain is irreducible, because there are many ways to transform an irreducible Markov chain to another Markov chain with a required stationary distribution. Below is a simplified version adopt from [8]. This algorithm is often called **Metropolis algorithm**.

Metropolis Algorithm: For a finite state space S and transition relation $\{N(x) | x \in S\}$, let $N = \max_{x \in S} |N(x)|$. Let M be any number such that $M > N$. Consider a Markov chain where

$$P_{x,y} = \begin{cases} 1/M & \text{if } x \neq y, y \in N(x) \\ 0 & \text{if } x \neq y, y \notin N(x), \\ 1 - N(x)/M & \text{if } x = y \end{cases}$$

If this chain is irreducible and aperiodic, then the stationary distribution is the uniform distribution.

2.5 Mixing-time and Rapidly Mixing

Although the Metropolis algorithm guarantees the chain's stationary distribution is uniform distribution. It may need infinity steps to converge to stationary distribution. Hence we need to measure how quickly the chain converges, which is important in algorithm analysis. We define the variation distance first.

Definiton 5 The **variation distance** between two distributions D_1 and D_2 on a countable state space S is given by

$$\|D_1 - D_2\| = \frac{1}{2} \sum_{x \in S} |D_1(x) - D_2(x)|$$

Definiton 6 Let π be the stationary distribution of a Markov chain with state space S . Let p_x^t represent the distribution of the state of the chain starting at state x after t steps. We define

$$\Delta_x(t) = \|p_x^t - \pi\|; \Delta(t) = \max_{x \in S} \Delta_x(t).$$

We also define

$$\tau_x(\epsilon) = \max\{t : \Delta_x(t) \leq \epsilon\}$$

$$\tau(\epsilon) = \max_{x \in S} \tau_x(\epsilon).$$

That is, $\Delta_x(t)$ is the variation distance between the stationary distribution and p_x^t , and $\Delta(t)$ is the maximum value over all states x . And $\tau_x(\epsilon)$ is the first step t which the variation distance between p_x^t and the stationary distribution is less than ϵ , $\tau(\epsilon)$ is the maximum value over all states x .

When $\tau(\epsilon)$ is considered as a function of ϵ , it can be called the **mixing-time** of the Markov chain. A chain is called **rapidly mixing** if $\tau(\epsilon)$ is polynomial in $\log(1/\epsilon)$ and the size of the problem.

2.6 Techniques for Analyzing Mixing-time

Because this mixing-time is measurement of how fast do Markov chain converge, but the exact mixing-time is hard to analyze. There are many ways to bind the mixing-time. We describe two techniques

Definiton 7 A **coupling** of a Markov chain M_t with state space S is a Markov chain $Z_t = (X_t, Y_t)$ on the state space $S \times S$ such that:

$$\Pr(X_{t+1} = x' | Z_t = (x, y)) = \Pr(M_{t+1} = x' | M_t = x)$$

$$\Pr(X_{t+1} = y' | Z_t = (x, y)) = \Pr(M_{t+1} = y' | M_t = y)$$

That is, a coupling consists of two copies of the Markov chain M that runs concurrently. Once X and Y reach the same state, they are said to have coupled.

Lemma 2 *Let $Z_t = (X_t, Y_t)$ be a coupling for a Markov chain M on a state space S . Suppose that there exists a T such that for every $x, y \in S$, $\Pr(X_t \neq Y_t | X_0 = x, Y_0 = y) < \epsilon$, then $\tau(\epsilon) \leq T$. This lemma is often called coupling lemma.*

Another technique is the extension of coupling lemma, which is called path coupling[2]. The idea behind path coupling is to consider state pairs which are close to each other. Suppose that we have shown that the expected change in distance is decreasing for all of these pairs. For any other states $x, y \in S$, we can find the mixing-time by linearity of expectation.

Theorem 1 *Let d be an integer valued metric defined on $S \times S$, which value range from 0 to B . If for any $(x, y) \in S \times S$, $P_{x,y}^0 > 0$, then $d(x, y) = 1$. If there exists $\beta < 1$ and $E[d(x_{t+1}, y_{t+1})] \leq \beta d(x_t, y_t)$, for all $(x_t, y_t) \in S \times S$, then*

$$\tau(\epsilon) \leq \frac{\ln(B\epsilon^{-1})}{1 - \beta}$$

3 Algorithm

The problem what we want to solve is that random generate the 2-connected graph. Because the algorithm is based on the MCMC method, the distribution is not really uniform distribution. Hence the problem is modified slightly.

Problem definition: *Given a positive integer n . Design a random generation algorithm for 2-connected simple graph with vertex size n , which sample distribution approximates uniform distribution. All isomorphic graphs are considered different.*

We design the Markov chain as follows. The state space is all n -vertices 2-connected graph. If two states x, y , whose edge set differs only one edge, then $P_{x,y} = 1/n^2$. The probability $P_{x,x}$ for all x is $1 - \sum_{i \neq x} P_{x,i}$. Other probabilities $P_{x,y} = 0$. Below is the algorithm we proposed based on the Markov chain.

Theorem 2 *Algorithm 1's output is a 2-connected graph.*

Algorithm 1 Sample 2-connected graph

Input: A positive integer n

Output: A 2-connected graph with n vertices

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1: Let graph  $G = (V, E)$  be a cycle with  $n$  vertices
2: for  $i = 0$  to  $n^2 \log n$  do
3:   Randomly choose a pair  $(u, v)$ ,  $u, v \in V$  and
   a random bit  $b$ 
4:   if  $b = 0$  and  $(u, v) \in E$  and  $(V, E - (u, v))$ 
   is a 2-connected graph then
5:      $E \leftarrow E - (u, v)$ 
6:   else if  $b = 1$  and  $u \neq v$  and  $(u, v) \notin E$ 
   then
7:      $E \leftarrow E \cup (u, v)$ 
8:   end if
9: end for
10: Output  $G$ 

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Proof: The technique that we used for proving correctness is loop invariant.

Loop invariant: During step 2 to 9, G is 2-connected graph.

Initialization Before the main loop is executed, G is initialized as a n vertices graph which is cycle. A cycle is a 2-connected graph obviously.

Maintenance Inside the main loop, step 5 and 7 may change the graph. In step 5, when the if condition in step 4 is true, then it promises that G after step 5 is 2-connected. On the other hand, 2-connected property is preserved under adding edge. Hence step 5 and 7 do not violate the loop invariant.

Termination The loop terminates when the loop executes r times. It has no impact on G . Hence the step 9 will output a 2-connected graph G . It means that this algorithm is correct. $\square$

Next, we want to prove that this algorithm's output distribution approximates uniform distribution. Because of the algorithm is based on the MCMC method and Metropolis algorithm. We only need to prove the Markov chain used in the algorithm 1 is aperiodic and irreducible.

Lemma 3 *The Markov chain used in the algorithm 1 is aperiodic.*

Proof: By definition, if every state x , $P_{x,x} > 0$, then the Markov chain is aperiodic. For any state x , there are at most $\binom{n}{2}$ transitions to other state, since the graph is simple. Hence the probability $P_{x,x} = 1 - \sum_{i \neq x} P_{x,i} \leq 1 - (n^2 - n)/n^2 > 0$. $\square$

Lemma 4 *The Markov chain used in the algorithm 1 is irreducible.*

Proof: Since the 2-connected property is preserved under adding edge, the Markov chain is symmetry. It means that if two states x, y if $P_{x,y} > 0$, then $P_{y,x} > 0$. Hence we prove that the chain is irreducible by for any two states x, y is accessible each other. For any two states x, y , which $x = (V, E_1)$ and $y = (V, E_2)$. The graph $G' = (V, E_1 \cup E_2)$ will be a 2-connected graph, because the 2-connected property is preserved under adding edge. Hence there is a state $z = G'$, such that x is accessible to z and y is accessible to z . $\square$

Theorem 3 *The output distribution of algorithm 1 approximates uniform distribution.*

Proof: According to lemma 1 and lemma 2, the chain is aperiodic and irreducible. By definition, the Markov chain's stationary distribution is uniform distribution. Hence the algorithm 1's output distribution approximates uniform distribution. $\square$

4 Mixing-time

We design a coupling for this Markov chain M . Let $Z_t = (X_t, Y_t)$ on the state space $S \times S$. If (u, v) and random bit b is chosen in X , then we choose (u, v) and b in Y . Although X and Y is not independent, this is still a proper coupling of Markov chain M .

In order to use path coupling technique, we define the distance metric d . For two n -vertices graph $G_1 = (V, E_1), G_2 = (V, E_2)$, $d(G_1, G_2) = (|E_1 - E_2| + |E_2 - E_1|)$. Because any n -vertices simple graph has at most $N = \binom{n}{2}$ edges, the maximum value for d is N .

Before we prove the main theorem, we define the **D-edge** to be the edge of a 2-connected graph G which removal results the graph is no more 2-connected. If G is a 2-connected graph, then $D(G)$ is the number of D-edge in G .

Main Theorem: Algorithms 1 is rapidly mixing

Proof: Suppose that $X_t = (V, E_x), Y_t = (V, E_y)$ and $d(X_t, Y_t) = 1$. For this coupling approach, we classify the edge (u, v) which is chosen randomly into three classes. Suppose that $|E_x| = |E_y| - 1$, the $|E_x| = |E_y| + 1$ case is symmetry.

1. $(u, v) \notin E_x \cup E_y$: the probability is $(n^2 - |E_x \cup E_y|)/n^2$, no matter what random bit is, this case does not change the distance.

2. $(u, v) \in (E_x \cup E_y) - (E_x \cap E_y)$: the probability is $d(X_t, Y_t)/n^2$, and no matter what random bit is, this case decrease the distance by one.
3. $(u, v) \in E_x \cap E_y$ and (u, v) is D-edge in one of E_x and E_y but not both: this case increase distance by one.
4. Other edges $(u, v) \in E_x \cap E_y$: the probability of case 3 and 4 is $(E_x \cap E_y)/n^2$, no matter what random bit is, this case does not change the distance.

Then we calculate the expectation

$$\begin{aligned} & E[d(X_{t+1}, Y_{t+1}) | d(X_t, Y_t) = 1] \\ &= \Pr(\text{case1}) + \Pr(\text{case4}) + 2 * \Pr(\text{case3}) - \Pr(\text{case2}) \\ &= 1 - n^{-2} + \Pr(\text{case3}) \end{aligned}$$

By the path coupling theorem, if we can prove that for any state x, y , when $d(x, y) = 1$ implies $\Pr(\text{case3}) < n^{-2}$, then we can conclude that the mixing-time is smaller than $O(n^2 \log n/\epsilon)$.

In order to calculate the $\Pr(\text{case3})$, we have some observations. First, if one of x, y is 4-connected graph, then $\Pr(\text{case3}) = 0$, since y must be 3-connected, and x, y have no D-edge. Next, when n is sufficiently large then almost always all n -vertices graphs are 4-connected. Integrating these two observations, if $\Pr(\kappa(G_{0.5}) < 4) = O(n^{-3})$ then the $\Pr(\text{case3}) \leq n^{-3}$.

$$\begin{aligned} \Pr(\text{case3}) &\leq \Pr(\kappa(G_{0.5}) < 4) \\ &\leq O(n^{-3}) \\ &= O(n^{-3}) \end{aligned}$$

Eventually, we want to prove this claim

Claim 1 $\Pr(\kappa(G_{0.5}) < 4) = O(2^{-n})$

For any graph G , if G is not 4-connected, then G has a separator set S , $|S| \leq 3$. $V(G) - S$ is the disjoint union of two non-empty sets, W_1 and W_2 , $|W_1| \leq |W_2|$, such that no edge between W_1 and W_2 . We denote the $G_{0.5}$ has a separator with size 3 with $|W_1| = r$ as A_r . Then

$$\begin{aligned} \Pr(\bigcup_r A_r) &\leq \sum_{r=1}^{\lfloor (n-3)/2 \rfloor} \binom{n}{r} \binom{n-r}{3} 2^{-r(n-r-3)} \\ &\leq \sum_{r=1}^{\lfloor (n-3)/2 \rfloor} \left(\frac{en}{r}\right)^r n^3 2^{-r(n/2-3)} \\ &\leq \sum_{r=1}^{\lfloor (n-3)/2 \rfloor} n^3 \left(\frac{en}{r}\right)^r 2^{(-n+3)r} \end{aligned}$$

$$\begin{aligned} &\leq \sum_{r=1}^{\lfloor (n-3)/2 \rfloor} n^3 2^{-nr} \\ &= O(n^{-3}) \end{aligned}$$

□

5 Conclusion and Future Work

In this paper, we described an algorithm that randomly sample 2-connected graph approximate uniformly. We also analyze the mixing-time roughly, and then prove it is almost rapidly mixing. In the future, we can still analyze the mixing-time deeply. On the other hand we also can extend our idea to any k-connected graph or some other complicated graph classes.

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